



A Practical Approach to Power, Effect Size and Sample Size & Introduction to PASS (Power Analysis and Sample Size software)

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Sample Size Considerations

- **Descriptive**
 - Gathering initial data in unknown area (e.g., feasibility studies)
 - Confidence intervals based on sample size
- **“Coverage”**
 - Ensuring all “types” of subjects are included
 - Get representative sample for each “type” (e.g., strata)
- **Hypothesis Testing**
 - Power/Sample Sizes based on Type 1 and 2 errors

Data analysis

- **Research objectives**
 - **Reduce bedsores**
- **Research questions**
 - **Does protocol A work better than protocol B for reducing bedsores?**
- **Status Quo – Initial Assumption (Null hypothesis)**
 - **Protocol A and B produce same % bedsores**
- **Where you want to go - Conclusion you'd like to reach (Alternative hypothesis)**
 - **Protocol A and B results in different % bedsores (one better than other)**

Framework for assessing the evidence

- **Threshold for assessing that outcome is real [SIGNIFICANCE]**
 - What is the level at which you want to be sure that your result was not obtained by just pure chance?
 - Usually set at 1-5% level of chance, i.e. 95-99% confidence in results
- **How much “risk” you’re willing to live with**
 - What is the level of “risk” assumed that you missed a real outcome [Beta = Type 2 error] usually 10-20%
 - So $POWER = 1 - \beta = 80\% - 90\%$
- **BOTH critical to determine BEFORE the study is performed and any data collected**

Framework for assessing the evidence

- Once you determine the level of confidence you want to achieve and the amount of risk you're willing to live with...
- How large an outcome do you expect/hope to see? To achieve clinical relevance? To change practice? [EFFECT SIZE]
- How large a SAMPLE SIZE is needed (given all of the above)
 - Time
 - Money/resource constraints

4 integrated keys

TRADEOFFS

SIGNIFICANCE
set BEFORE –
confidence
level desired for
conclusions?

Significance

Power

POWER set
BEFORE – level
of risk that you
missed
something

EFFECT SIZE -
determined from
data or
estimated or
stated from
previous
observations
and clinical
goals

Effect size

Sample
size

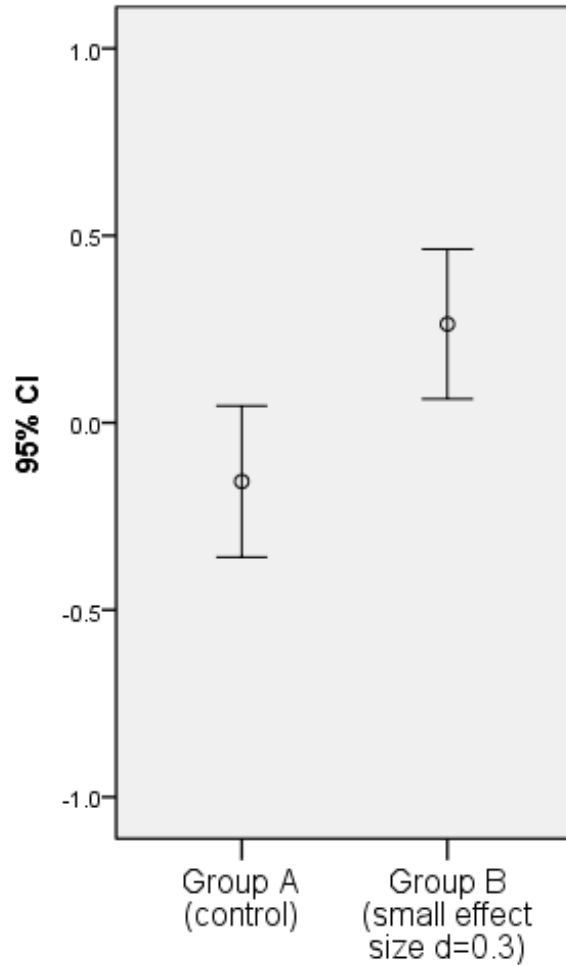
SAMPLE SIZE is
usually the MAIN
option you can
adjust within
budget and
study time
constraints.

What constitutes evidence?

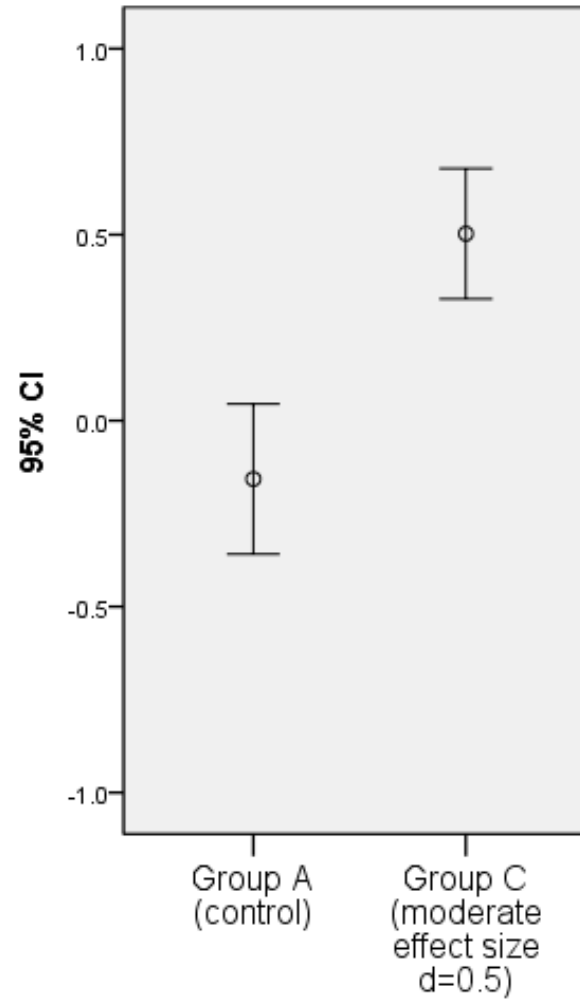
- **Big Sample Sizes in the 1000's**
 - Everything is statistically significant...
 - Is it clinically significant/relevant?
 - It is actionable?
- **Small Sample Sizes**
 - Nothing statistically significant
 - But were clinically significant changes seen?
- **Was the difference due to the “intervention”/”treatment”?
Or due to something else (that you measured or not)?**

Visualize effect size – ERROR BARS

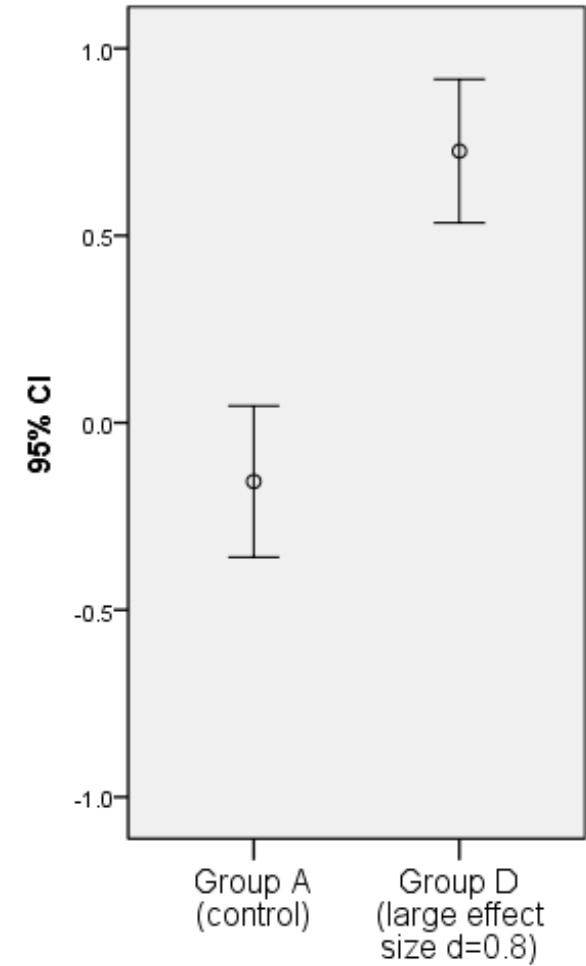
SMALL



MEDIUM



LARGE





Hypothesis Testing

- USA Legal System: “We ASSUME that a person is INNOCENT until PROVEN GUILTY.”
- Null Hypothesis (“Status Quo”)
 - H_0 : Person = Innocent
- Alternative Hypothesis (“What We’d Like to PROVE”)
 - H_a : Person = Guilty
- “Burden of Proof” is on Prosecution [There has to be ENOUGH EVIDENCE to REJECT Innocence and CONCLUDE GUILT.]
- Underlying Structure – we’d rather let a guilty person go free (beta, type 2) than send an innocent person to jail (alpha, type 1)... more later...



From a Stats Point of View

- **Null Hypothesis [H_0]:** The Hypothesis to be “Tested” – really this is what we want to reject in favor of the **Alternative Hypothesis**
- **Alternative Hypothesis [H_a]:** The stated alternative to the null – really what we would like to “accept” and “conclude”
- **Hypothesis Test:** The test performed to make the decision to either “Accept H_a ” or “Not Reject H_0 ”
- **** NEVER ACCEPT THE NULL HYPOTHESIS **** Always State that “there was not enough evidence to reject the null hypothesis – this does NOT mean the Null Hypothesis is True. [Just because the prosecution could not prove their case does not mean the defendant is innocent.]



POWER – Some initial terms and definitions

- H_0 is the Null Hypothesis (want to reject this to conclude H_a – the alternative hypothesis) **** Research Design ****
- H_a is the Alternative Hypothesis (ideally this is the conclusion you want to reach)
- If statistical significance is found ($p\text{-value} < \alpha$), “You reject the null and accept the alternative hypothesis.”
- If the test is NOT significant you state that “You can not reject the null hypothesis.” *US Legal System*
- **YOU NEVER ACCEPT THE NULL HYPOTHESIS!!**



Types of “Error”

		<u>“Truth”</u>	
		“H ₀ is True”	“H _a is True (H ₀ is false)”
<u>“Result”</u>	Fail to Reject H ₀	☒ 1-α Confidence Level	β Type II error
	Reject H ₀ , Conclude H _a	α Type 1 error	☒ POWER 1-β

- **Type I Error (α Significance Level):** Rejecting the Null Hypothesis when it is in fact true
- **Type II Error (β):** Not Rejecting the Null Hypothesis when it is in fact false.



Power – The logic

		<u>“Truth”</u>	
		“H ₀ is True”	“H _a is True (H ₀ is false)”
<u>“Result”</u>	Fail to Reject H ₀		β Type II error
	Reject H ₀ , Conclude H _a	α Type 1 error	 POWER

- **POWER = 1 – β**

Our focus is usually here – controlling Type I (α).
All tied together – Beta (β) depends upon Alpha (α)

- A statistical test's **POWER** is the probability of rejecting a false statistical null hypothesis (re: Statistica Textbook)

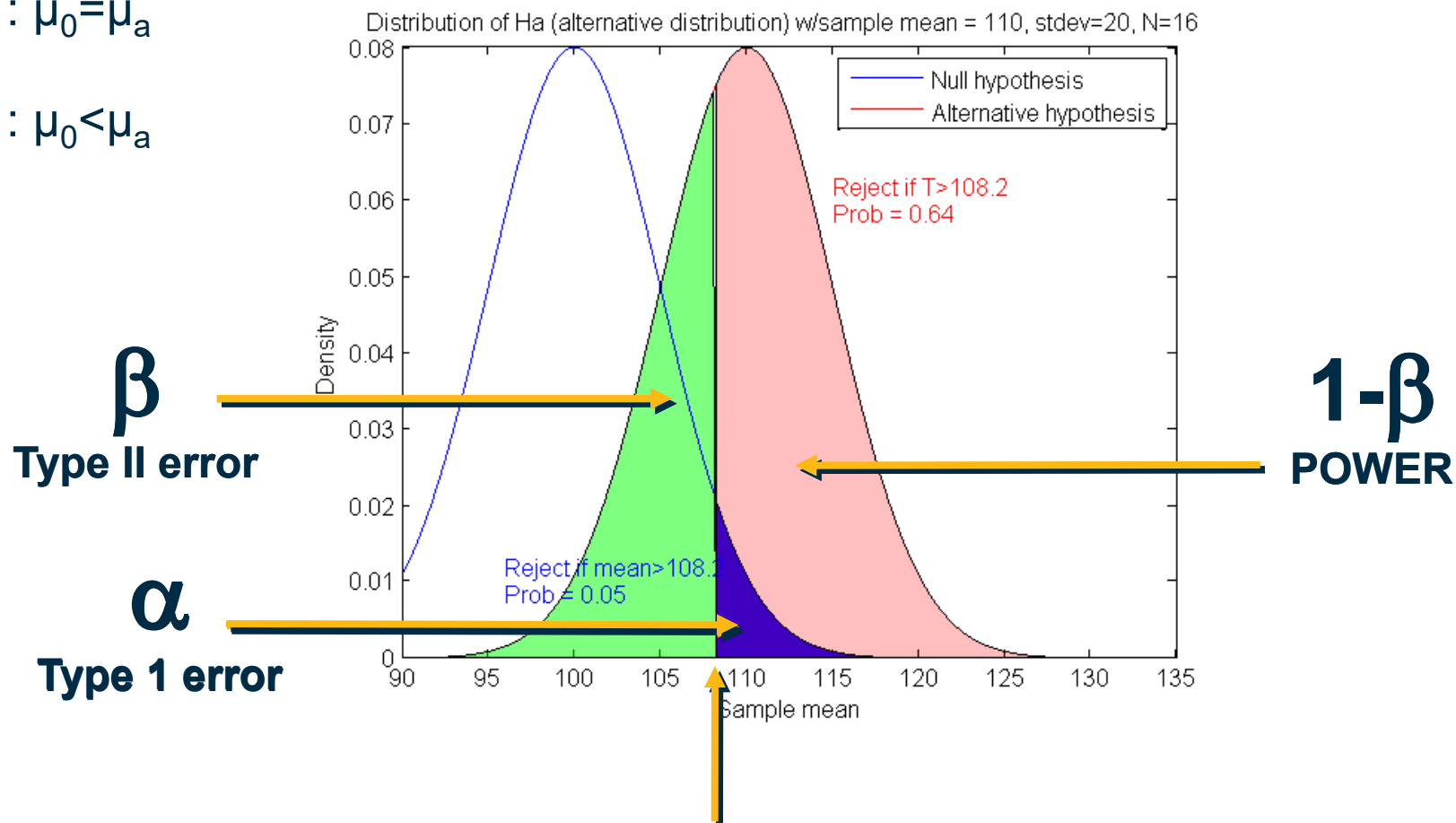


Power – The pictures

~~DISTRIBUTION UNDER THE NULL HYPOTHESIS~~
~~DISTRIBUTION UNDER THE ALTERNATIVE HYPOTHESIS~~

$$H_0 : \mu_0 = \mu_a$$

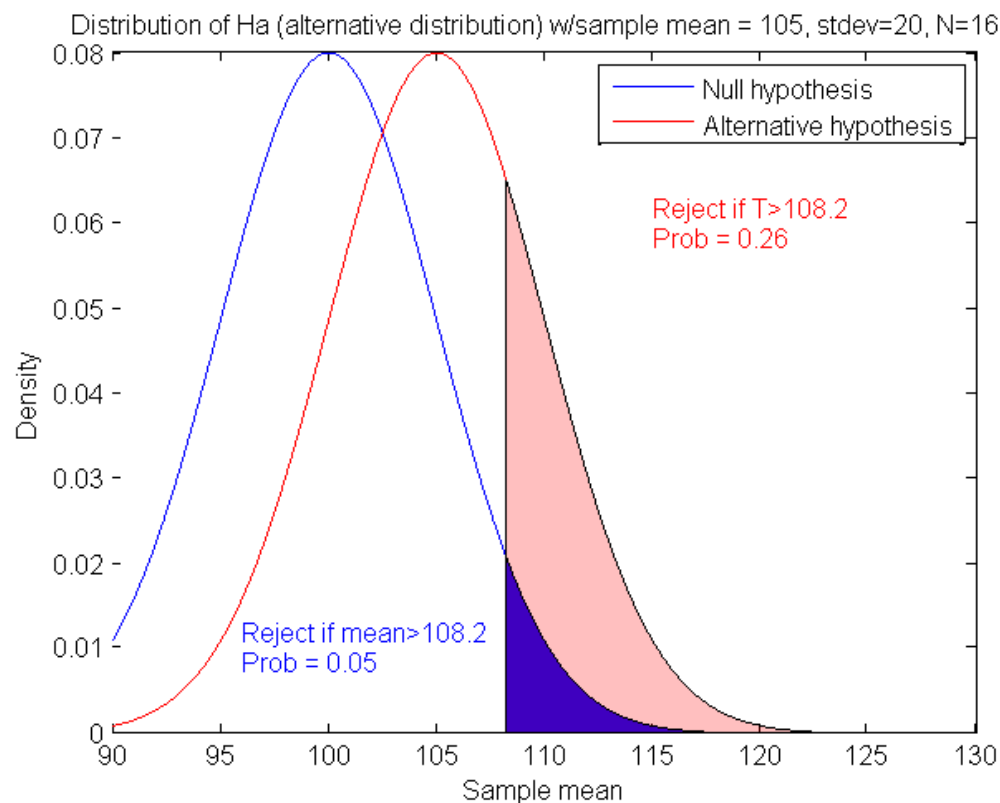
$$H_a : \mu_0 < \mu_a$$



Notice that everything depends on the “critical value” = 108.2, which depends on alpha (α)



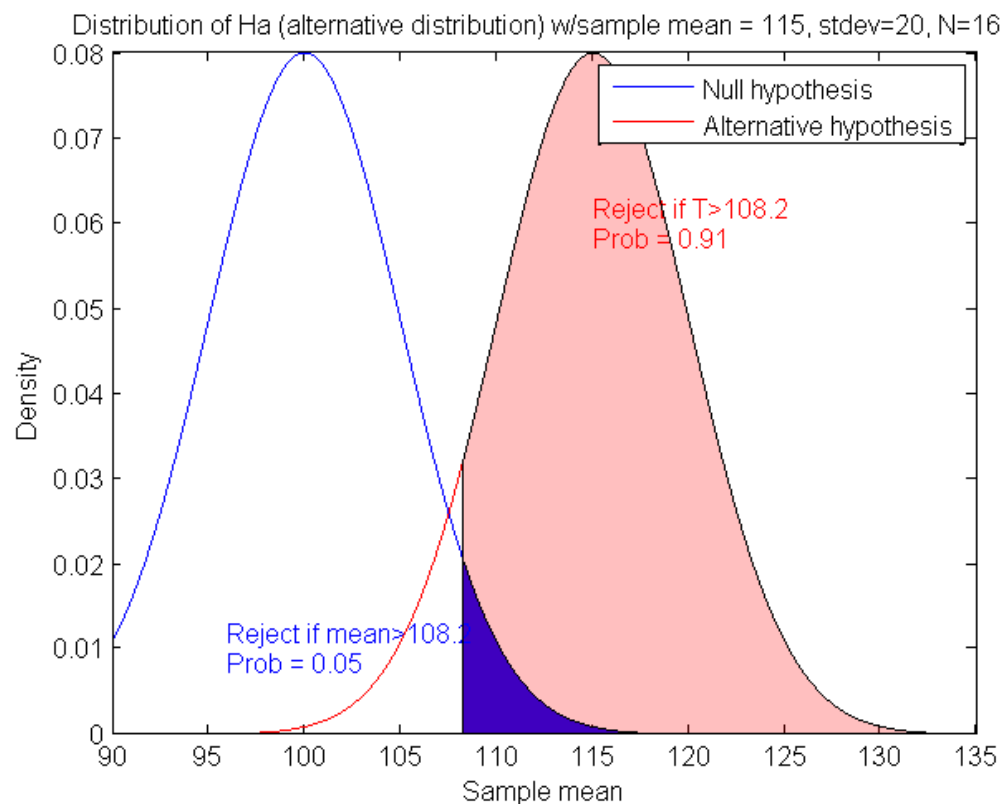
What if $\mu_0=100$, $\mu_a=105$ ($\sigma_x=20$ & $N=16$) ?
Closer together – will power increase or decrease?



Power = 0.26 (previously was 0.64)
 $\mu_a - \mu_0 = 5$ (previous difference = 10)



What if $\mu_0=100$, $\mu_a=115$ ($\sigma_x=20$ & $N=16$) ?



Power = 0.91 (was 0.64)
 $\mu_a - \mu_0 = 15$ (previous difference = 10)

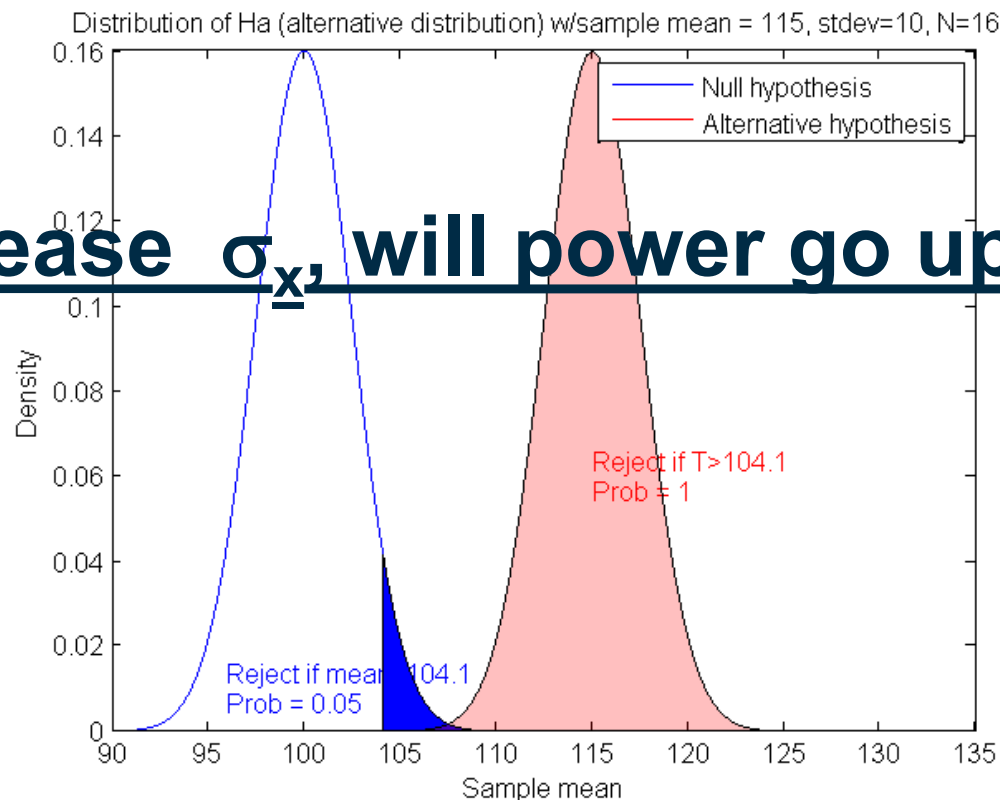


What if $\mu_0=100$, $\mu_a=105$ ($\sigma_{\bar{x}}=10$, $N=16$) ? Power = 0.64

What if $\mu_0=100$, $\mu_a=110$ ($\sigma_{\bar{x}}=10$, $N=16$) ? Power = 0.99

What if $\mu_0=100$, $\mu_a=115$ ($\sigma_{\bar{x}}=10$, $N=16$) ? Power = 1.00

If we decrease $\sigma_{\bar{x}}$, will power go up or down?



The “effect” of increasing the difference between the means increases power, i.e. $\mu_a - \mu_0$ is related to “effect size” of this test.



Summary so far

This is for N=16, what if we have
a different N (sample size)?

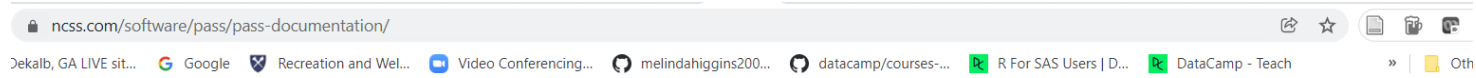
μ_0	μ_a	$\mu_a - \mu_0$	σ_x	<u>Power</u>
100	105	5	20	0.26
100	110	10	20	0.64
100	115	15	20	0.91
100	105	5	10	0.64
100	110	10	10	0.99
100	115	15	10	1.00



PASS Power Analysis Software

PASS – go to <http://www.ncss.com/pass.html>

- download Trial for FREE, Annual License \$695] but
- documentation FREE,
<https://www.ncss.com/software/pass/pass-documentation/>



PASS Documentation

Use the links below to load individual chapters from the PASS statistical software training documentation in PDF format. The chapters correspond to the procedures available in PASS. Each chapter generally has an introduction to the topic, technical details including power and sample size calculation details, explanations for the procedure options, examples, and procedure validation examples. Each of these chapters is also available through the PASS help system when running the software.

Jump to topic:

- Quick Start
- Introduction
- Bayesian Approaches
- Cluster-Randomized
- Conditional Power
- Confidence Intervals
- Correlation
- Design of Experiments
- Equivalence
- GEE
- Group-Sequential
- Means
- Method Comparison
- Microarray
- Mixed Models
- Non-Inferiority
- Nonparametric
- Normality
- Pilot Studies
- Post-Marketing Surveillance
- Proportions
- Quality Control
- Rates and Counts
- Regression

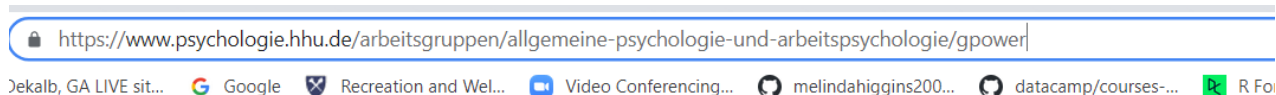
Introduction

- License Agreement
- The PASS Home Window
- The Procedure Window
- The Output Window
- Introduction to Power Analysis
- Power Analysis of Proportions
- Power Analysis of Means



G*Power Software - FREE

<https://www.psychologie.hhu.de/arbeitsgruppen/allgemeine-psychologie-und-arbeitspsychologie/gpower>



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Help

[Download the G*Power manual \(PDF\)](#)

[Download the Short Tutorial of G*Power \(PDF\) written for G*Power 2 but still useful as an introduction](#)

For more help, see the papers about G*Power in the References section below.



G*Power Software - FREE

<https://www.psychologie.hhu.de/arbeitsgruppen/allgemeine-psychologie-und-arbeitspsychologie/gpower>

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🔗 Download G*Power 3.1.9.6 for Mac OS X 10.7 to 12 (about 2 MB).



Other Power Software

- **Power and Precision (BioStat) – go to <http://www.power-analysis.com/>**
- **StudySize 3.0 – go to <http://www.studysize.com/> [14-day FREE trial; \$95 for StudySize 3.0 Academic]**
- **“Russ Lenth's power and sample-size page” – go to <http://www.math.uiowa.edu/~rlenth/Power/> [access JAVA applets online or download for FREE]**
- **For a large list of available “Power, Sample Size and Experimental Design Calculations” – go to <http://statpages.org/#Power>**



Hands-on exercises

Effect Size (for t-test)

[Cohen, J. “Statistical Power Analysis for the Behavioral Sciences (2nd ed)” (1988)]

- **Example for t-test:**

- **Cohen’s d**

$$d = \frac{|\mu_1 - \mu_2|}{\sigma}$$

where μ_1 and μ_2 are population means
and σ is the population standard deviation
(assumed to be equal for both groups)

- **d = 0.2 “small”**

- meaning that the difference in the means is 20% as large as the “common” standard deviation. So, if the standard deviation in a particular measure is +/- 10, a small effect size would be a difference of 2.

- **d = 0.5 “medium”**

- **d = 0.8 “large”**

- **Use these numbers as guidelines for your study. Calculate an approximate ratio of the expected difference in means divided by a conservative (largest) estimate for the standard deviation.**



Numeric Results for Two-Sample T-Test

Null Hypothesis: Mean1=Mean2. Alternative Hypothesis: Mean1<>Mean2

The standard deviations were assumed to be unknown and equal.

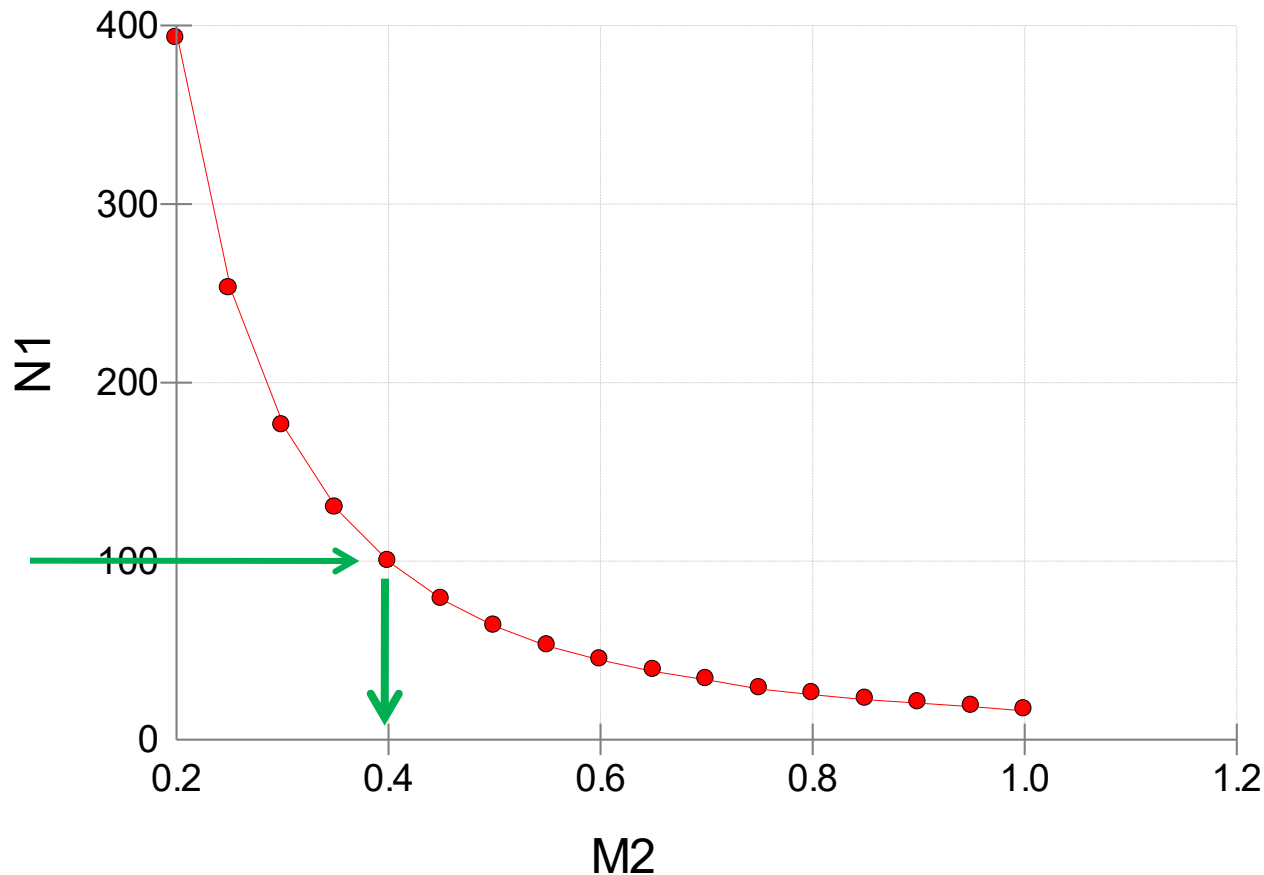
Power	N1	N2	Allocation		Beta	Mean1	Mean2	S1	S2
			Ratio	Alpha					
0.80003	1570	1570	1.000	0.05000	0.19997	0.000	0.100	1.000	1.000
0.80044	393	393	1.000	0.05000	0.19956	0.000	0.200	1.000	1.000
0.80138	176	176	1.000	0.05000	0.19862	0.000	0.300	1.000	1.000
0.80365	100	100	1.000	0.05000	0.19635	0.000	0.400	1.000	1.000
0.80146	64	64	1.000	0.05000	0.19854	0.000	0.500	1.000	1.000
0.80370	45	45	1.000	0.05000	0.19630	0.000	0.600	1.000	1.000
0.81165	34	34	1.000	0.05000	0.18835	0.000	0.700	1.000	1.000
0.80749	26	26	1.000	0.05000	0.19251	0.000	0.800	1.000	1.000
0.81211	21	21	1.000	0.05000	0.18789	0.000	0.900	1.000	1.000
0.80704	17	17	1.000	0.05000	0.19296	0.000	1.000	1.000	1.000

Since the standard deviation of each group = 1, and Mean1 (for the 1st group) is set = to 0, then Mean2 is the effect size of interest.

This table presents results for effect sizes from 0.1 to 1.0.

Thus, you need 393 per group (786 total) to “detect” an effect size of 0.2 and, you need 64 per group (128 total) to “detect” an effect size of 0.5

N1 vs M2 with M1=0.000 S1=1.000 S2=1.000
Alpha=0.050 Power=0.807 N2=N1 2-Sided T



Remember that $N1=N2$, so the sample sizes represented here have to be doubled to estimate total sample sizes needed. So, for a sample size of 100+100 (200 total), we can “detect” an effect size of 0.4 (M2).



G*Power: Independent T-Test

G*Power:

Family:
T-Tests

Test:
Means Diff between 2
independent means

Type:
A Priori compute sample size

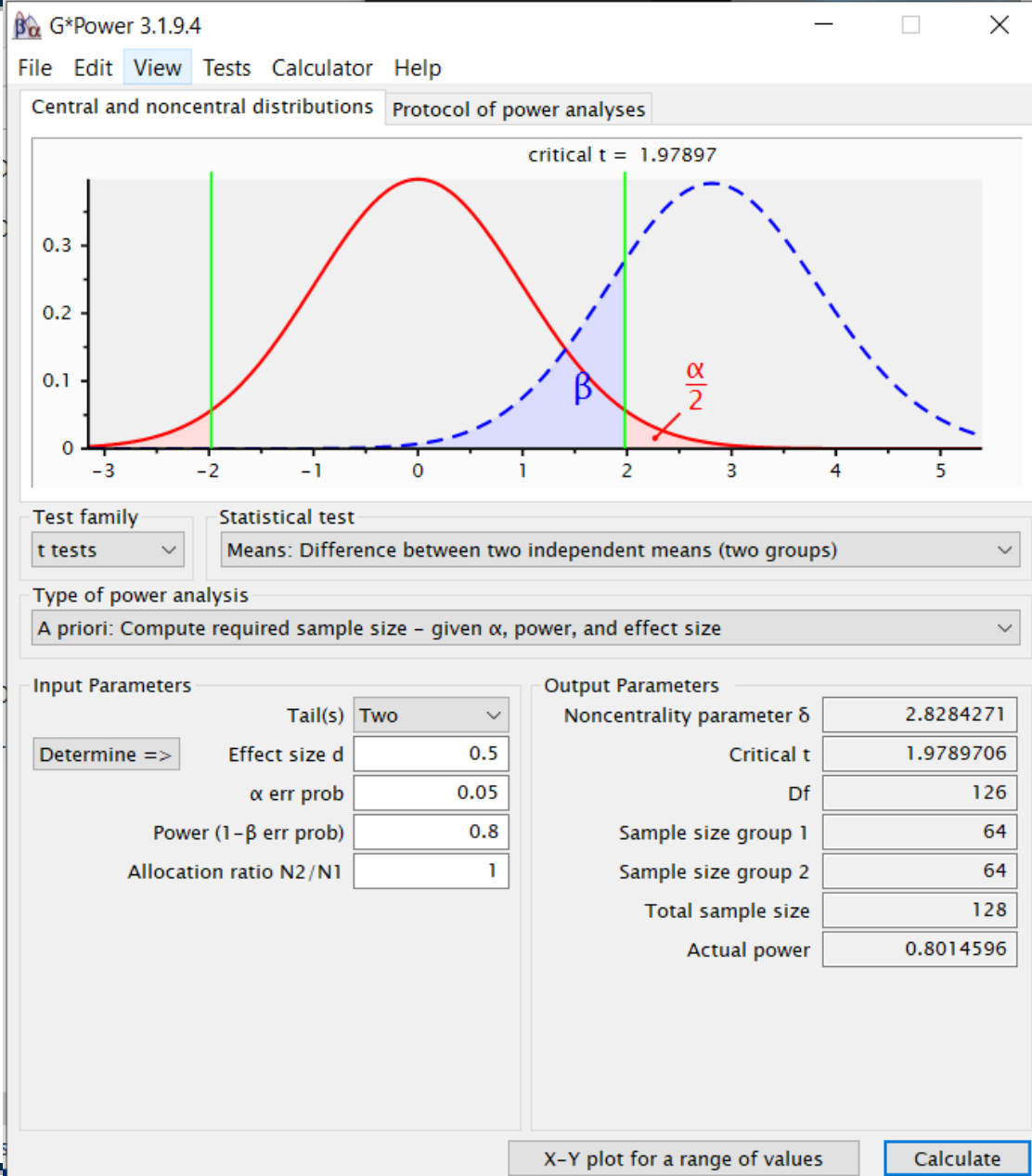
Tails: Two

Effect Size d:
(small) $d=0.2$
(mod) $d=0.5$
(large) $d=0.8$

Power: 0.80

Click CALCULATE

Also Try X-Y Plot for Range



Power, Effect Size and Sample Size



G*Power: Independent T-Test

G*Power:

Family:
T-Tests

Test:
Means Diff between 2
independent means

Type:
A Priori compute sample size

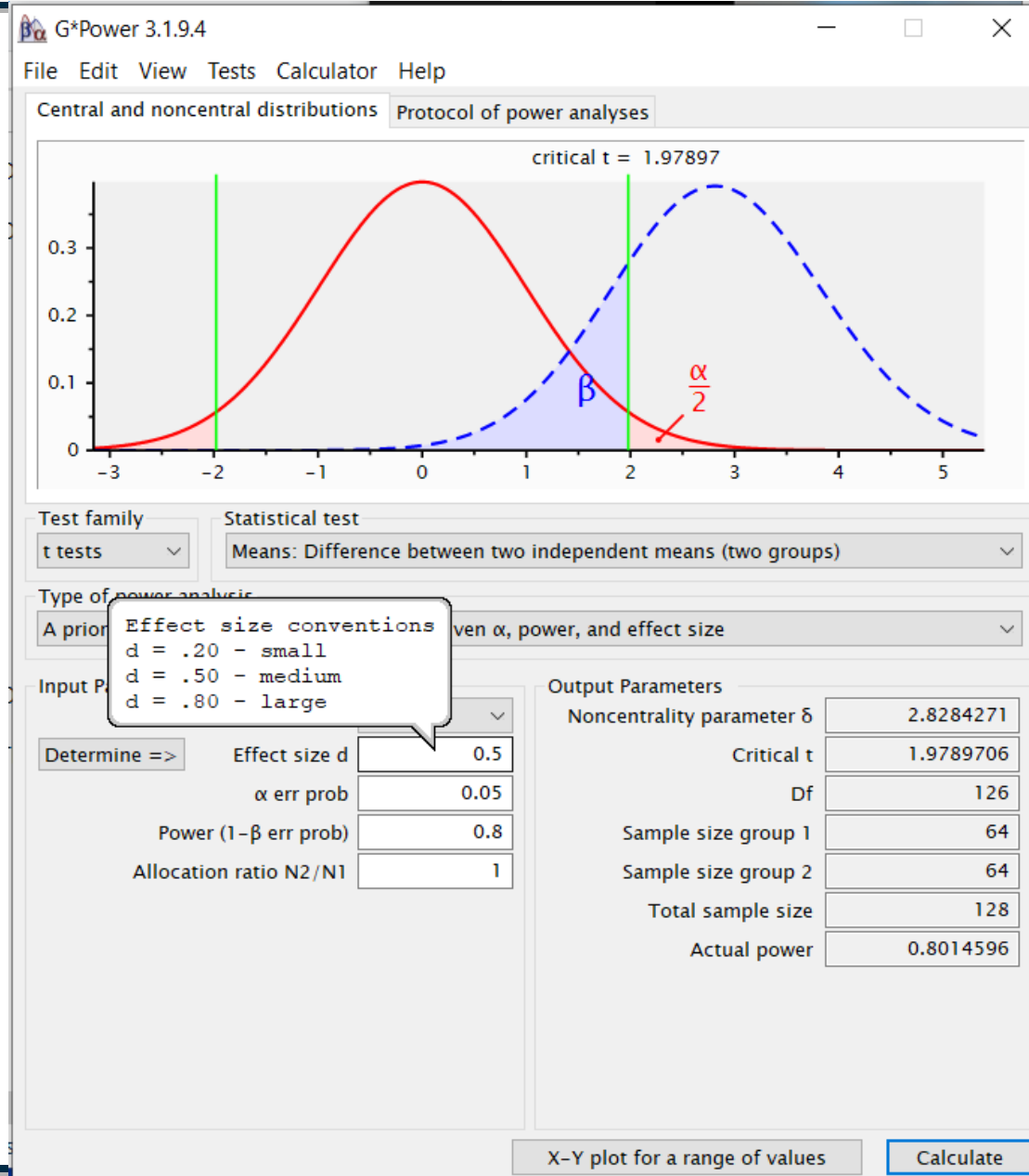
Tails: Two

Effect Size d:
(small) $d=0.2$
(mod) $d=0.5$
(large) $d=0.8$

Power: 0.80

Click CALCULATE

Also Try X-Y Plot for Range



Power, Effect Size and Sample Size



Test	Effect Size	Sample Size for 5% Significance Level and 80% Power
2-group Independent T-test	d = 0.2 (small) d = 0.5 (mod) d = 0.8 (large)	394*2 = 788 64*2 = 128 26*2 = 52

Effect Size for Chi-Square

- Effect Size for Chi-Square test

- Cohen's "w"

$$w = \sqrt{\frac{\chi^2}{N}} \quad \text{or} \quad Nw^2 = \chi^2$$

where N is the total sample size

- $w = 0.1$ "small"

- indicating that expected Chi-square will be 1% (0.01) of the total sample size (N); $w^2 = (0.1)^2 = 0.01$

- $w = 0.3$ "medium"

- $w = 0.5$ "large"

- Use these numbers as guidelines for your study. Typically estimate a Chi-square from previous studies or pilot data.



G*Power:

Family:

χ^2 (chi-square) tests

Test:

Goodness-of-fit tests

Type:

A Priori compute sample size

Effect Size w:

(small) $w=0.1$

(mod) $w=0.3$

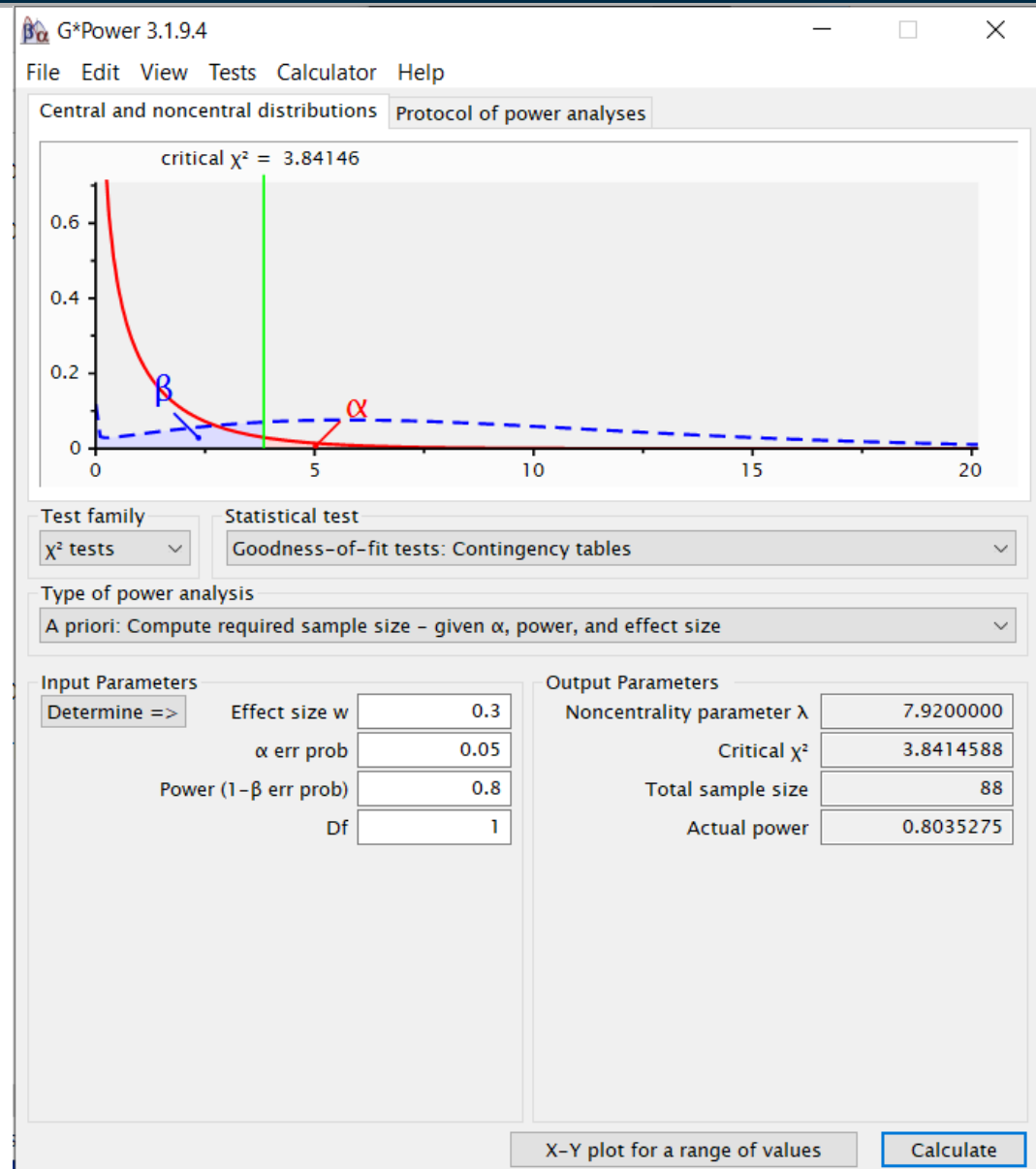
(large) $w=0.5$

Power: 0.80

Df = 1 ((r-1)*(c-1))

Click CALCULATE

Also Try X-Y Plot for Range





G*Power:

Family:

χ^2 (chi-square) tests

Test:

Goodness-of-fit tests

Type:

A Priori compute sample size

Effect Size w:

(small) $w=0.1$

(mod) $w=0.3$

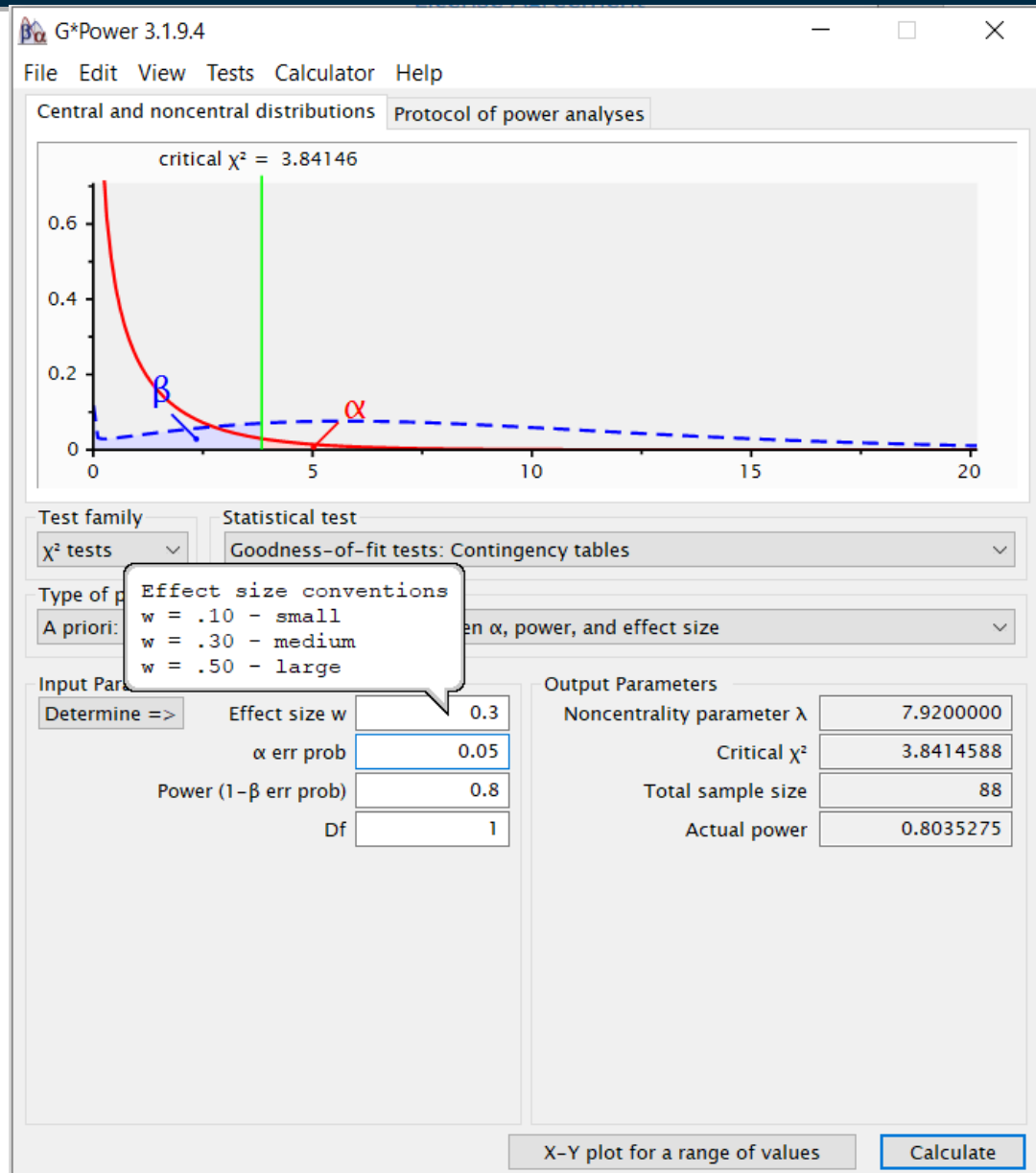
(large) $w=0.5$

Power: 0.80

Df = 1 ((r-1)*(c-1))

Click CALCULATE

Also Try X-Y Plot for Range





Test	Effect Size	Sample Size for 5% Significance Level and 80% Power
Chi-square test	(small)	
	$\omega = 0.1$ (<i>df=1, for 2x2 table</i>)	785
	$\omega = 0.1$ (<i>df=2, for 2x3 table</i>)	964
	$\omega = 0.1$ (<i>df=3, for 2x4 table</i>)	1091
	(moderate)	
	$\omega = 0.3$ (<i>df=1, for 2x2 table</i>)	88
	$\omega = 0.3$ (<i>df=2, for 2x3 table</i>)	108
	$\omega = 0.3$ (<i>df=3, for 2x4 table</i>)	122
	(large)	
	$\omega = 0.5$ (<i>df=1, for 2x2 table</i>)	32
	$\omega = 0.5$ (<i>df=2, for 2x3 table</i>)	39
	$\omega = 0.5$ (<i>df=3, for 2x4 table</i>)	44



Effect Size (for Correlation/Regression)

[Cohen, J. "Statistical Power Analysis for the Behavioral Sciences (2nd ed)" (1988)]

- Correlation (r and r²) IS an Effect Size

• "r"

$$r = \sqrt{\frac{\left[\sum (x_i - \bar{x})^2 \right] \left[\sum (y_i - \bar{y})^2 \right]}{\left[\sum (x_i - \bar{x})(y_i - \bar{y}) \right]}}$$

- $r = 0.1$ "small" $r^2 = 0.01$
 - $r = 0.3$ "medium" $r^2 = 0.09$
 - $r = 0.5$ "large" $r^2 = 0.25$
- Use these numbers as guidelines for your study.



G*Power:

Family:

T-Tests

Test:

Correlation: Pt Biserial

Type:

A Priori compute sample size

Tails: Two

Effect Size ρ : [same as r]

(small) $\rho = 0.1$

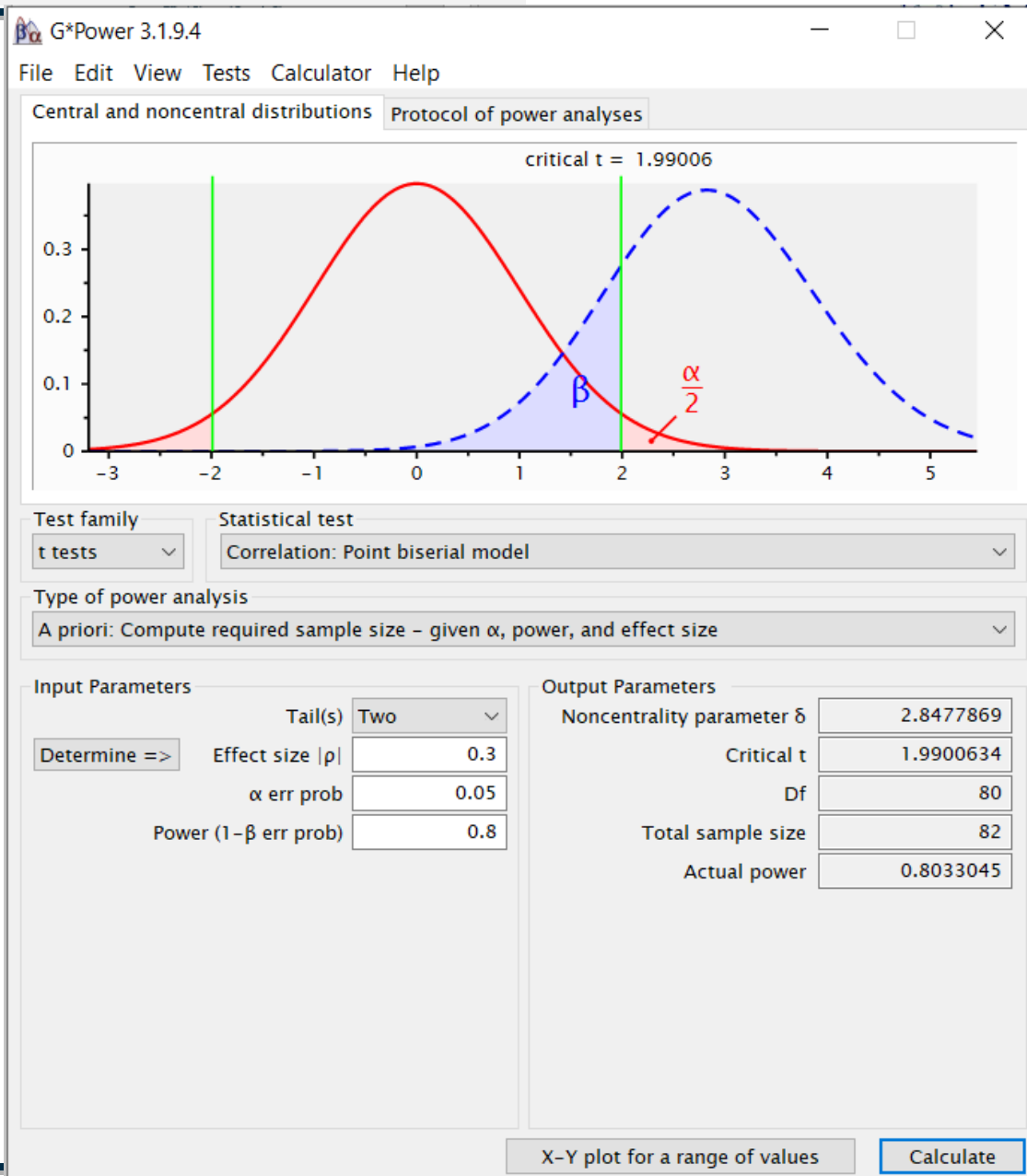
(mod) $\rho = 0.3$

(large) $\rho = 0.5$

Power: 0.80

Click CALCULATE

Also Try X-Y Plot for Range



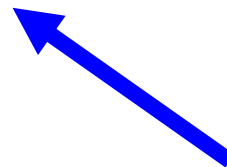


Test	Effect Size	Sample Size for 5% Significance Level and 80% Power
Pearson's Correlation	$r = 0.1$ (sm) ($r^2=0.01$) $r = 0.3$ (mod) ($r^2=0.09$) $r = 0.5$ (large) ($r^2=0.25$)	779 82 26



(Simple) Linear Regression

$$Y = \beta_0 + \beta_1 * X$$



Slope for 1 predictor, 1 outcome = correlation (r)



G*Power:

Family:
T-Tests

Test:
Linear Bivariate Reg:
One group, size of slope

Type:
A Priori compute sample size

Tails: Two

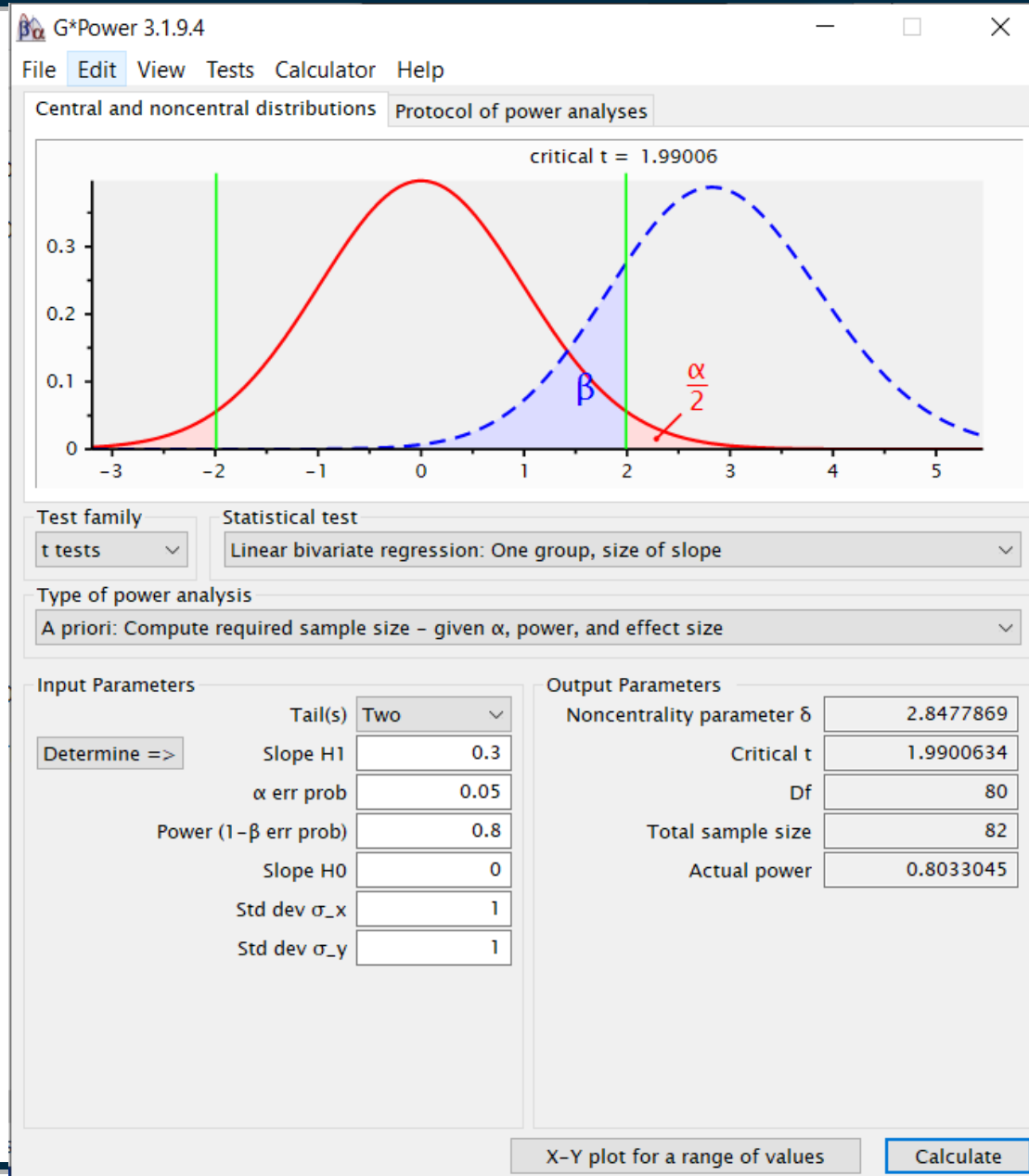
Effect Size Slope H1 (r):
(small) $r=0.1$
(mod) $r=0.3$
(large) $r=0.5$

Power: 0.80

Std dev for x and y = 1

Click CALCULATE

Also Try X-Y Plot for Range



Effect Size and Sample Size



G*Power:

Family:

F-Tests

Test:

Linear multiple reg:

Fixed model R² deviation

Type:

A Priori compute sample size

Effect Size f^2 :

(small) $f^2 = 0.02$

(mod) $f^2 = 0.15$

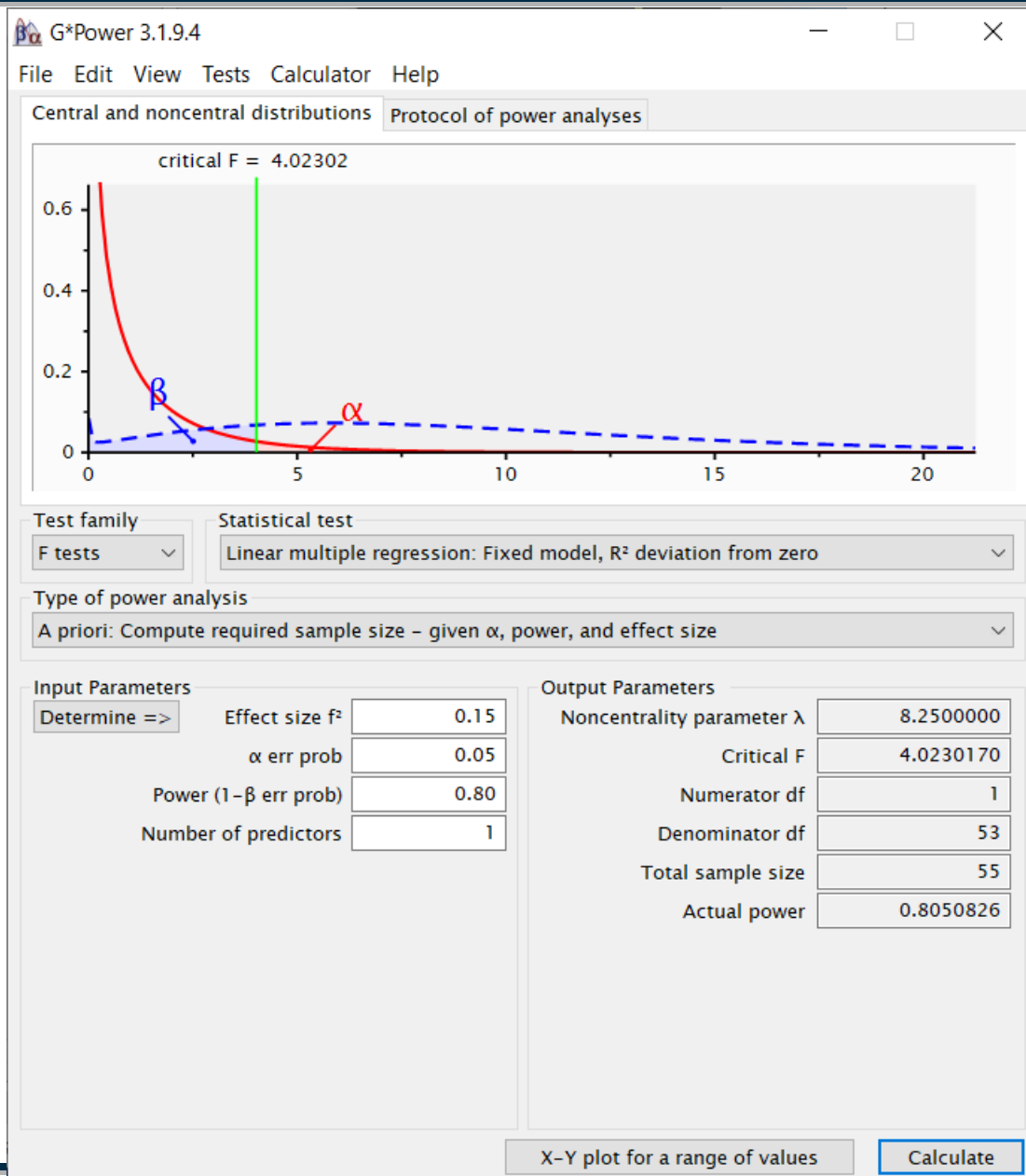
(large) $f^2 = 0.35$

Power: 0.80

Number predictors: 1

Click CALCULATE

Also Try X-Y Plot for Range





G*Power:

Family:

F-Tests

Test:

Linear multiple reg:

Fixed model R² deviation

Type:

A Priori compute sample size

Effect Size f^2 :

(small) $f^2 = 0.02$

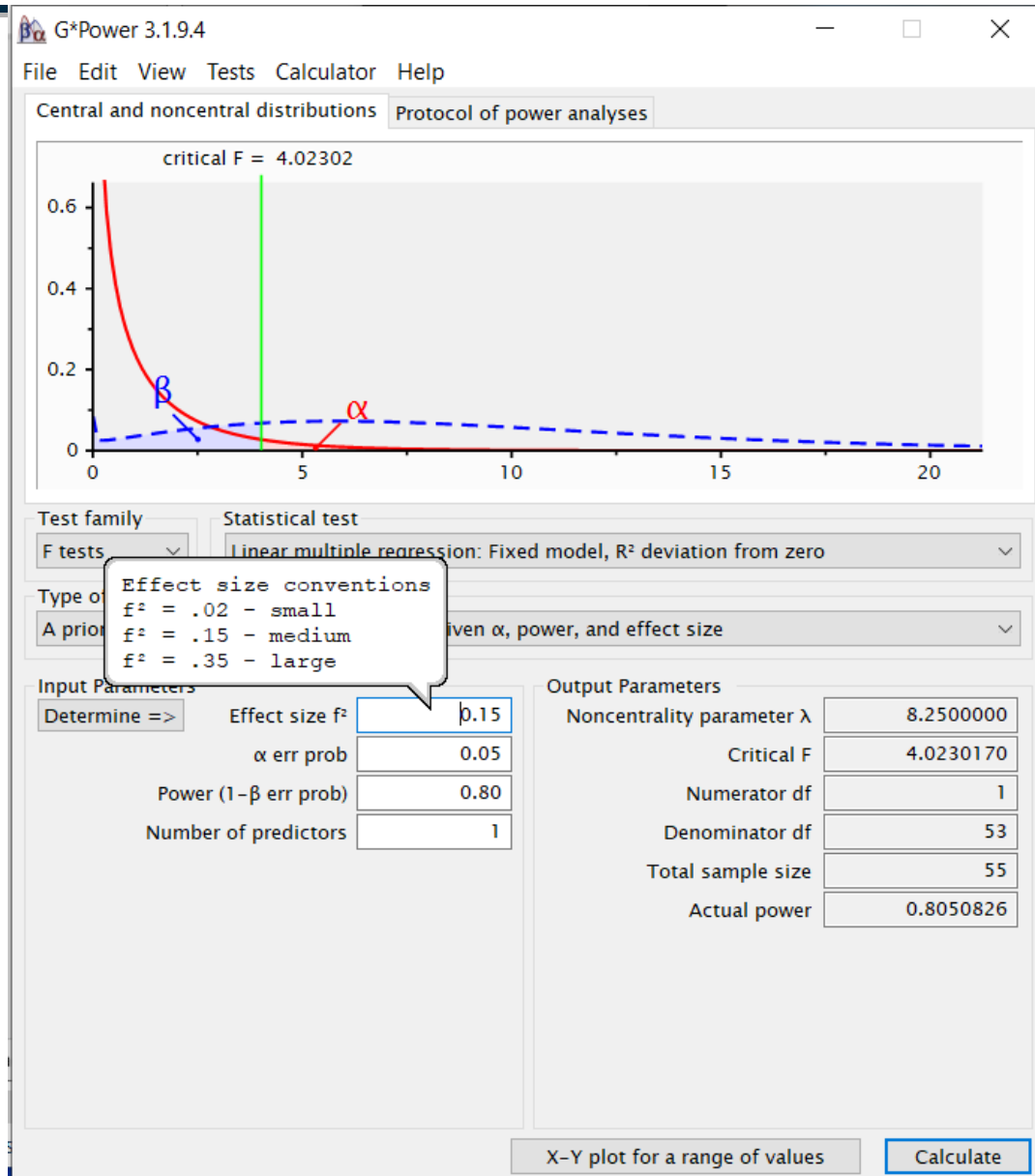
(mod) $f^2 = 0.15$

(large) $f^2 = 0.35$

Power: 0.80

Click CALCULATE

Also Try X-Y Plot for Range





Test	Effect Size	Sample Size for 5% Significance Level and 80% Power
Simple Linear Regression (1 predictor)	$f^2 = 0.02$ (sm)	395
	$f^2 = 0.0989$ ($r=0.3$, $r^2=0.09$)	82
	$f^2 = 0.15$ (mod)	55
	$f^2 = 0.35$ (large)	25



https://www.psychometrica.de/effect_size.html - see #14

14. Transformation of the effect sizes d , r , f , Odds Ratio, η^2 and Common Language Effect Size (CLES)

Please choose the effect size, you want to transform, in the drop-down menu. Specify the magnitude of the effect size in the text field on the right side of the drop-down menu afterwards. The transformation is done according to Cohen (1988), Rosenthal (1994, S. 239), Borenstein, Hedges, Higgins, and Rothstein (2009; transformation of d in Odds Ratios) and Dunlap (1994; transformation in CLES).

$$\begin{aligned} r &= 0.3 \\ r^2 &= 0.09 = \eta^2 \\ f &= 0.3145 \\ f^2 &= 0.0989 \end{aligned}$$

Effect Size	r	0.3
d	0.629	
r	0.3	
η^2	0.09	
f	0.3145	
Odds Ratio	3.1294	
Common Language Effect Size CLES	59.7 %	
Number Needed to Treat (NNT)	2.9112	





Sample Size recommendations

- Number of model parameters (p) (usually number of predictors) $< n$
 - Rules of thumb: typically assume moderate effect sizes at 5% level of significance (α = Type I error = 0.05) and 80% power (Type II error = 0.20)
 - $N \Rightarrow p \times 10$ – sample size is 10x larger than number of predictors/parameters in model
 - Green (1991) for overall model fit: $N \Rightarrow 50 + 8p$ – for 5 predictors you need 90+ subjects
 - Green (1991) for testing individual predictors: $N \Rightarrow 104 + p$ – so for 5 predictors you need 109+ subjects
 - Khamis and Kepler (2010): $N \Rightarrow 20 + 5p$ – for 5 predictors you'd need 45 – consider this a minimum when power estimates are not feasible
- Need larger sample size when DV skewed, more measurement errors, smaller effect sizes



Power, Sample Size, Effect Size – G*Power

- Given effect sizes $f^2=0.02$ (small), 0.15 (moderate), 0.35 (large)
- Green (1991):
 - $N \Rightarrow (8/f^2) + (p-1)$
 - For $p=5$, $f^2=0.02$, $N \Rightarrow 404$
 - For $p=5$, $f^2=0.15$, $N \Rightarrow 57.3 \sim 58$
 - For $p=5$, $f^2=0.35$, $N \Rightarrow 26.9 \sim 27$
- Better yet – use software like G*Power



Linear Regression: Change in “r²”

$$Y = \beta_0 + \underbrace{\beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3}_{\text{controlling for 3 covariates}} + \underbrace{\beta_4 X_4 + \beta_5 X_5}_{\text{2 tested predictors}}$$

Suppose we want a final model with 5 predictors:

Where we are “controlling for” 3 covariates

And testing the significance of 2 predictors AFTER controlling for the first 3

You are interested in the CHANGE in r² for the 2 tested predictors



G*Power:

Family:

F-Tests

Test:

Linear multiple reg:

Fixed model R² increase

Type:

A Priori compute sample size

Effect Size f^2 :

(small) $f^2 = 0.02$

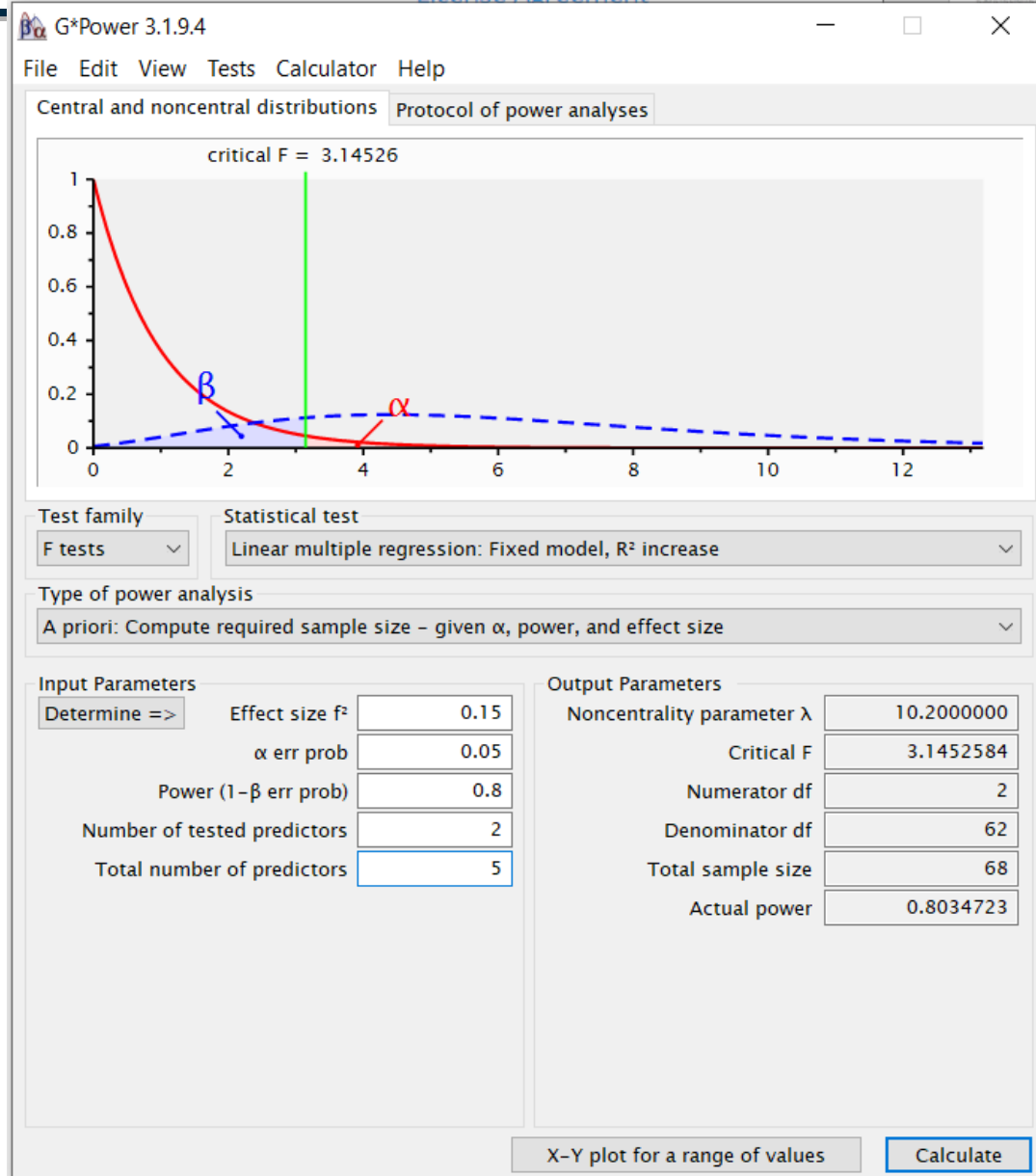
(mod) $f^2 = 0.15$

(large) $f^2 = 0.35$

Power: 0.80

Tested predictors: 2

TOTAL predictors: 5





Logistic Regression

14. Transformation of the effect sizes d , r , f , Odds Ratio, η^2 and Common Language Effect Size (CLES)

Please choose the effect size, you want to transform, in the drop-down menu. Specify the magnitude of the effect size in the text field on the right side of the drop-down menu afterwards. The transformation is done according to Cohen (1988), Rosenthal (1994, S. 239), Borenstein, Hedges, Higgins, and Rothstein (2009; transformation of d in Odds Ratios) and Dunlap (1994; transformation in CLES).

Effect Size	d	0.5
d		0.5
r		0.2425
η^2		0.0588
f		0.25
Odds Ratio		2.4766
Common Language Effect Size CLES		63.82 %
Number Needed to Treat (NNT)		3.6189



G*Power:

Family:

z Tests

Test:

Logistic Regression

Type:

A Priori compute sample size

Tails: Two

Odds Ratio: 2.5 (mod)

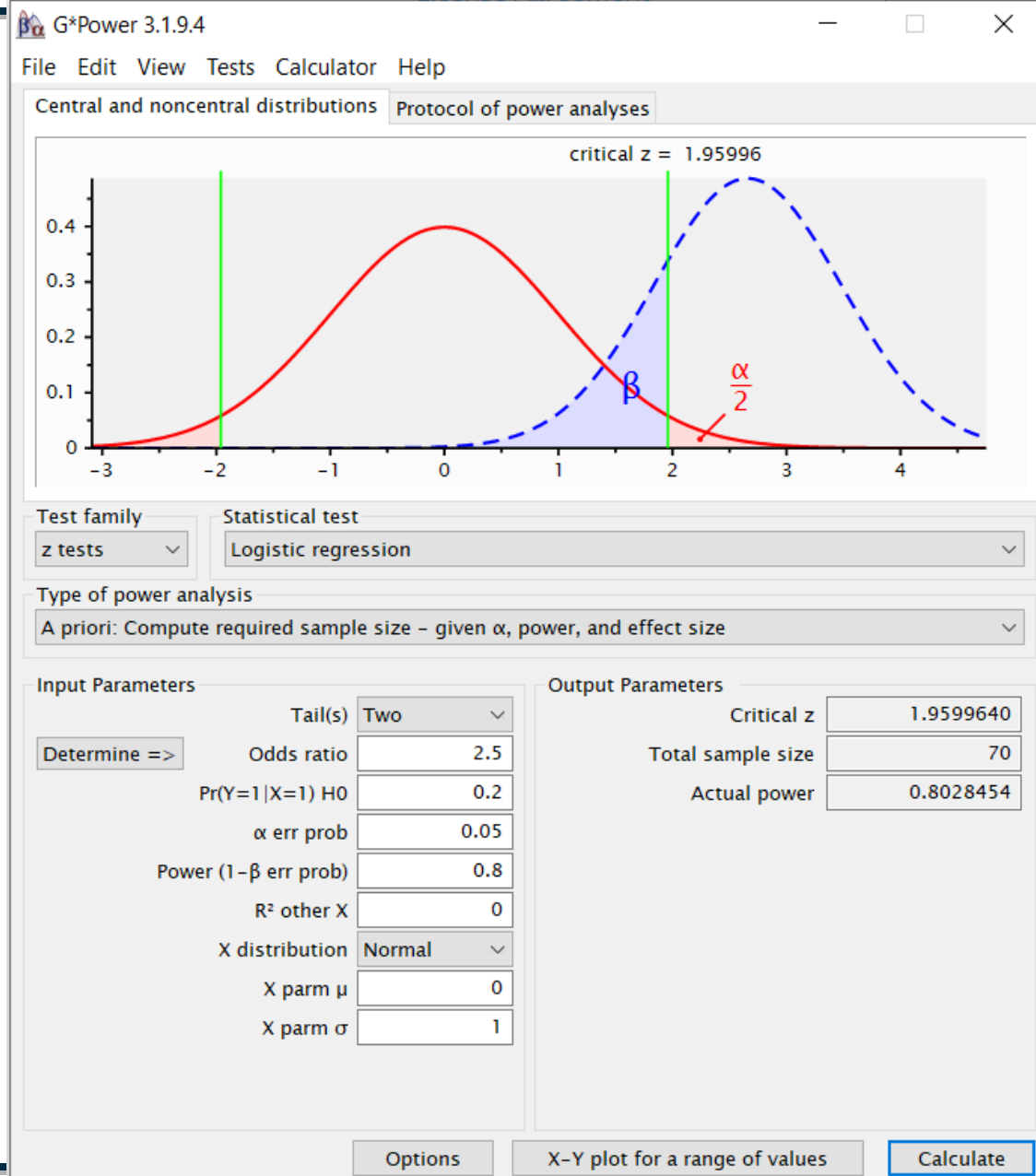
$\Pr(Y=1|X=1)H_0 = 0.2$

Power: 0.80

[keep defaults]

Click CALCULATE

Also Try X-Y Plot for Range





Test	Effect Size	Sample Size for 5% Significance Level and 80% Power
Logistic regression	$\Pr(Y=1 \mid X=1) H_0 = 0.2$ OR = 1.4 (~ d=0.2, sm) OR = 2.5 (~ d=0.5, mod) OR = 4.3 (~ d=0.8, lge)	442 70 36

Note: It is good to know the probability of (proportion of) your sample that will have your outcome of interest. The smaller this probability, the larger your sample needs to be.

$\Pr(Y=1 \mid X=1) H_0 = 0.2 = 20\%$ of your sample has your outcome of interest. This can be/will often be a limiting factor. E.g. for the sample size of 36, only 7 people would have the outcome of interest...



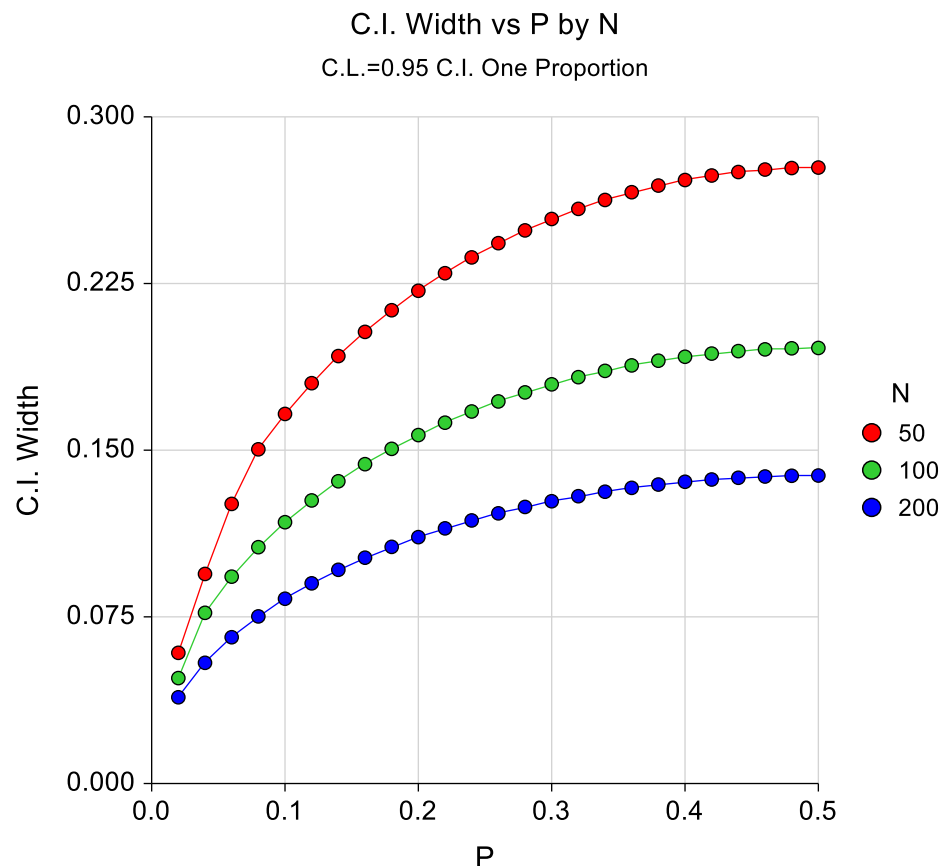
Summary/Points to Remembers

- **Never accept the null hypothesis.** Always state, “We can not reject the null hypothesis.”
- **Always design your “experiment” to reject the null hypothesis in order to accept the alternative (i.e. what you want to prove), given statistical significance is achieved.**
- **Identify the statistical test(s) appropriate for testing your hypothesis(es).**
- **Use the most conservative (largest) sample size estimate given your “effect size” of interest (i.e. clinically significant not statistically significant).**



Descriptive Statistics

Confidence Intervals for Proportions



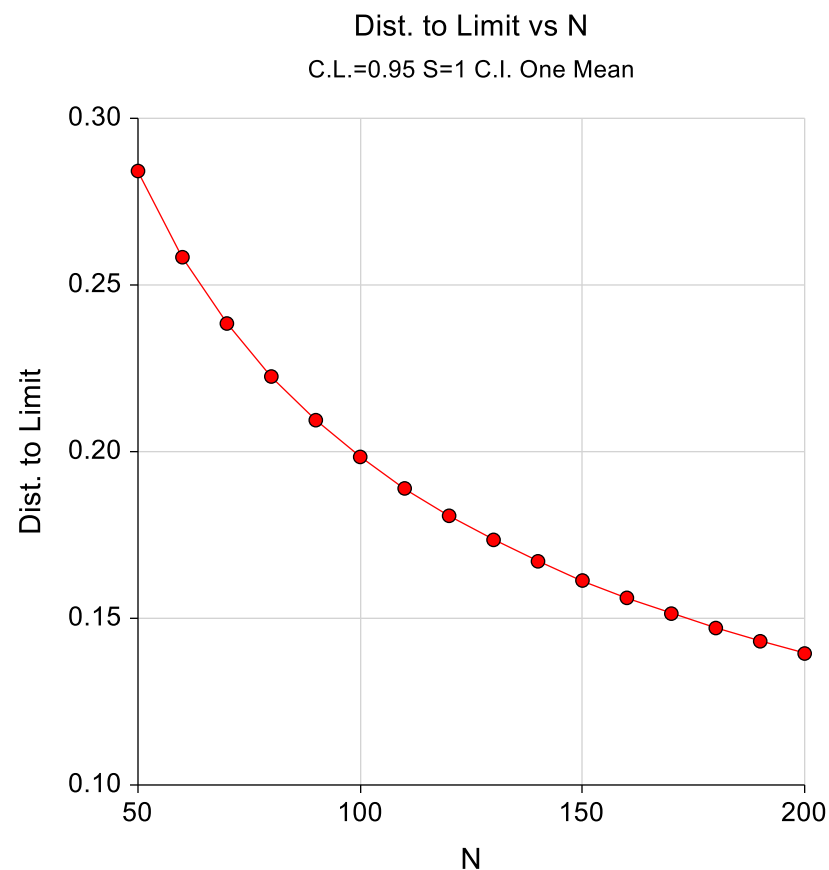
n	p	+/-	width
50	0.1	+/- 0.083	0.1663
100	0.1	+/- 0.059	0.1176
200	0.1	+/- 0.042	0.0832
50	0.5	+/- 0.139	0.2772
100	0.5	+/- 0.098	0.1960
200	0.5	+/- 0.069	0.1386

Performed in PASS 2024



Descriptive Statistics

Confidence Intervals for Means



n	mean	SD	+/-
50	0	1	+/- 0.284
80	0	1	+/- 0.223
110	0	1	+/- 0.189
140	0	1	+/- 0.167
170	0	1	+/- 0.151
200	0	1	+/- 0.139

Performed in PASS 2024



VIII. Statistical Resources and Contact Info

Contact

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